BIG MART DATA SALES ANALYSIS

In [1]:
```
import pandas as pd
import numpy as np
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
# IGNORE ALL WARNINGS
from warnings import filterwarnings
warnings=filterwarnings("ignore")
```

In [3]:
```
data=pd.read_csv('Train.csv')
```

In [4]:
```
data.head()
```

Out[4]:

	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Iden
0	FDA15	9.30	Low Fat	0.016047	Dairy	249.8092	OU
1	DRC01	5.92	Regular	0.019278	Soft Drinks	48.2692	OU
2	FDN15	17.50	Low Fat	0.016760	Meat	141.6180	OU
3	FDX07	19.20	Regular	0.000000	Fruits and Vegetables	182.0950	OU
4	NCD19	8.93	Low Fat	0.000000	Household	53.8614	OU

In [5]:
```
data.shape
```

Out[5]:
```
(8523, 12)
```

1). DATA CLEANING

In [6]:
```
# check the null values
data.isnull().sum()
```

Out[6]:
```
Item_Identifier                 0
Item_Weight                  1463
Item_Fat_Content                0
Item_Visibility                 0
Item_Type                       0
Item_MRP                        0
Outlet_Identifier               0
Outlet_Establishment_Year       0
Outlet_Size                  2410
Outlet_Location_Type            0
Outlet_Type                     0
Item_Outlet_Sales               0
dtype: int64
```

In [7]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8523 entries, 0 to 8522
Data columns (total 12 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   Item_Identifier            8523 non-null   object
 1   Item_Weight                7060 non-null   float64
 2   Item_Fat_Content           8523 non-null   object
 3   Item_Visibility            8523 non-null   float64
 4   Item_Type                  8523 non-null   object
 5   Item_MRP                   8523 non-null   float64
 6   Outlet_Identifier          8523 non-null   object
 7   Outlet_Establishment_Year  8523 non-null   int64
 8   Outlet_Size                6113 non-null   object
 9   Outlet_Location_Type       8523 non-null   object
 10  Outlet_Type                8523 non-null   object
 11  Item_Outlet_Sales          8523 non-null   float64
dtypes: float64(4), int64(1), object(7)
memory usage: 799.2+ KB
```

In [8]:
```
#check the mean,min,max in your data
data.describe()
```

Out[8]:

	Item_Weight	Item_Visibility	Item_MRP	Outlet_Establishment_Year	Item_Outlet_Sales
count	7060.000000	8523.000000	8523.000000	8523.000000	8523.000000
mean	12.857645	0.066132	140.992782	1997.831867	2181.288914
std	4.643456	0.051598	62.275067	8.371760	1706.499616
min	4.555000	0.000000	31.290000	1985.000000	33.290000
25%	8.773750	0.026989	93.826500	1987.000000	834.247400
50%	12.600000	0.053931	143.012800	1999.000000	1794.331000
75%	16.850000	0.094585	185.643700	2004.000000	3101.296400
max	21.350000	0.328391	266.888400	2009.000000	13086.964800

In [9]:
```
#check the mean,min,max in item weight
data['Item_Weight'].describe()
```

Out[9]:
```
count    7060.000000
mean       12.857645
std         4.643456
min         4.555000
25%         8.773750
50%        12.600000
75%        16.850000
max        21.350000
Name: Item_Weight, dtype: float64
```

Item weight is numerical value so we can fill it with mean imputation

In [10]:
```
data['Item_Weight'].fillna(data['Item_Weight'].mean(),inplace=True)
```

Outlet_Size is categorical value so we can fill it with mode imputation

In [11]:
```
data['Outlet_Size'].value_counts()
```

Out[11]:
```
Medium    2793
Small     2388
High       932
Name: Outlet_Size, dtype: int64
```

In [12]:
```
data['Outlet_Size'].fillna(data['Outlet_Size'].mode()[0],inplace=True)
```

In [13]:
```
#now you can check null values is filled
data.isnull().sum()
```

Out[13]:
```
Item_Identifier              0
Item_Weight                  0
Item_Fat_Content             0
Item_Visibility              0
Item_Type                    0
Item_MRP                     0
Outlet_Identifier            0
Outlet_Establishment_Year    0
Outlet_Size                  0
Outlet_Location_Type         0
Outlet_Type                  0
Item_Outlet_Sales            0
dtype: int64
```

view some basic details about dataset

In [14]:
```
#view unique items of Outlet_Establishment_Year
print(data['Outlet_Establishment_Year'].unique())
```

```
[1999 2009 1998 1987 1985 2002 2007 1997 2004]
```

In [15]:
```
#view unique items of Outlet_Size
print(data['Outlet_Size'].unique())
```

```
['Medium' 'High' 'Small']
```

In [16]:
```
#view unique items of Item_Type
data['Item_Type'].unique()
```

Out[16]:
```
array(['Dairy', 'Soft Drinks', 'Meat', 'Fruits and Vegetables',
       'Household', 'Baking Goods', 'Snack Foods', 'Frozen Foods',
       'Breakfast', 'Health and Hygiene', 'Hard Drinks', 'Canned',
       'Breads', 'Starchy Foods', 'Others', 'Seafood'], dtype=object)
```

In [17]:
```
#view unique items of Outlet_Type
data['Outlet_Type'].unique()
```

Out[17]:
```
array(['Supermarket Type1', 'Supermarket Type2', 'Grocery Store',
       'Supermarket Type3'], dtype=object)
```

In [18]:
```
#view unique items of Item_Fat_Content
data['Item_Fat_Content'].unique()
```

Out[18]:
```
array(['Low Fat', 'Regular', 'low fat', 'LF', 'reg'], dtype=object)
```

In [19]:
```
#change the names of Item_fat_content
data['Item_Fat_Content'].replace(['LF', 'low fat', 'reg'],
         ['Low Fat','Low Fat','Regular'],inplace=True)
```

In [20]:
```
#vie all columns names
data.columns
```

Out[20]:
```
Index(['Item_Identifier', 'Item_Weight', 'Item_Fat_Content', 'Item_Visibility',
       'Item_Type', 'Item_MRP', 'Outlet_Identifier',
       'Outlet_Establishment_Year', 'Outlet_Size', 'Outlet_Location_Type',
       'Outlet_Type', 'Item_Outlet_Sales'],
      dtype='object')
```

In [21]:
```
# drop unnecessary columns
data.drop(['Item_Identifier','Outlet_Location_Type'
          ,'Outlet_Identifier','Item_Visibility'],axis=1)
```

Out[21]:

	Item_Weight	Item_Fat_Content	Item_Type	Item_MRP	Outlet_Establishment_Year	Outlet_Size
0	9.300	Low Fat	Dairy	249.8092	1999	Medium
1	5.920	Regular	Soft Drinks	48.2692	2009	Medium
2	17.500	Low Fat	Meat	141.6180	1999	Medium
3	19.200	Regular	Fruits and Vegetables	182.0950	1998	Medium
4	8.930	Low Fat	Household	53.8614	1987	High
...	...	...	...	...	...	...
8518	6.865	Low Fat	Snack Foods	214.5218	1987	High
8519	8.380	Regular	Baking Goods	108.1570	2002	Medium
8520	10.600	Low Fat	Health and Hygiene	85.1224	2004	Small
8521	7.210	Regular	Snack Foods	103.1332	2009	Medium
8522	14.800	Low Fat	Soft Drinks	75.4670	1997	Small

8523 rows × 8 columns

some methods of eda

In [22]:
```
#using dtale
'''(pip install dtale) use this for install
import dtale
d= dtale.show(data)
dtale.instances()'''
#using pandas profiling
'''pip install
https://github.com/pandas-profiling/pandas-profiling/archive/master.zip
import pandas_profiling
profile= pandas_profiling.ProfileReport(data)'''
# using data pred
''''(pip install dataprep) use this for install
from dataprep.eda import create_report
create_report(data)'''
```

Out[22]:
```
"'(pip install dataprep) use this for install\nfrom dataprep.eda import create_
report\ncreate_report(data)"
```

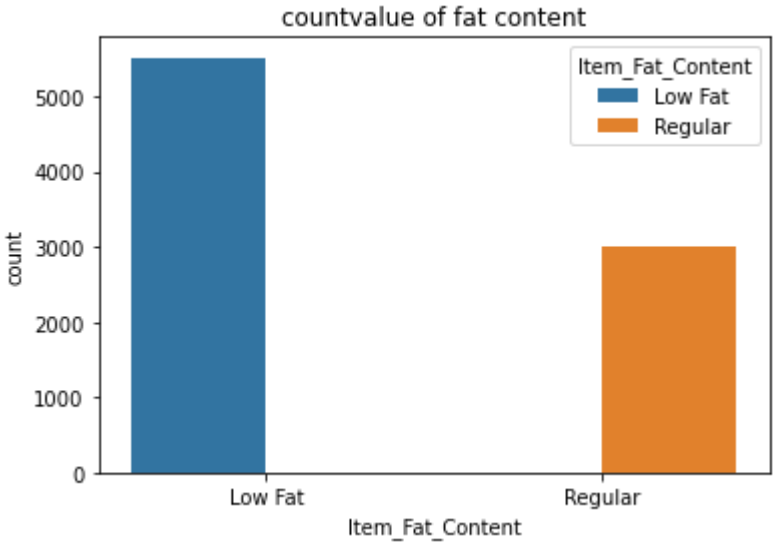
use sweetviz for show your report in html form

In [23]:
```
# (pip install sweetviz) use this for install
'''import sweetviz as sv
my_report = sv.analyze(data)
report = my_report.show_html()'''
```

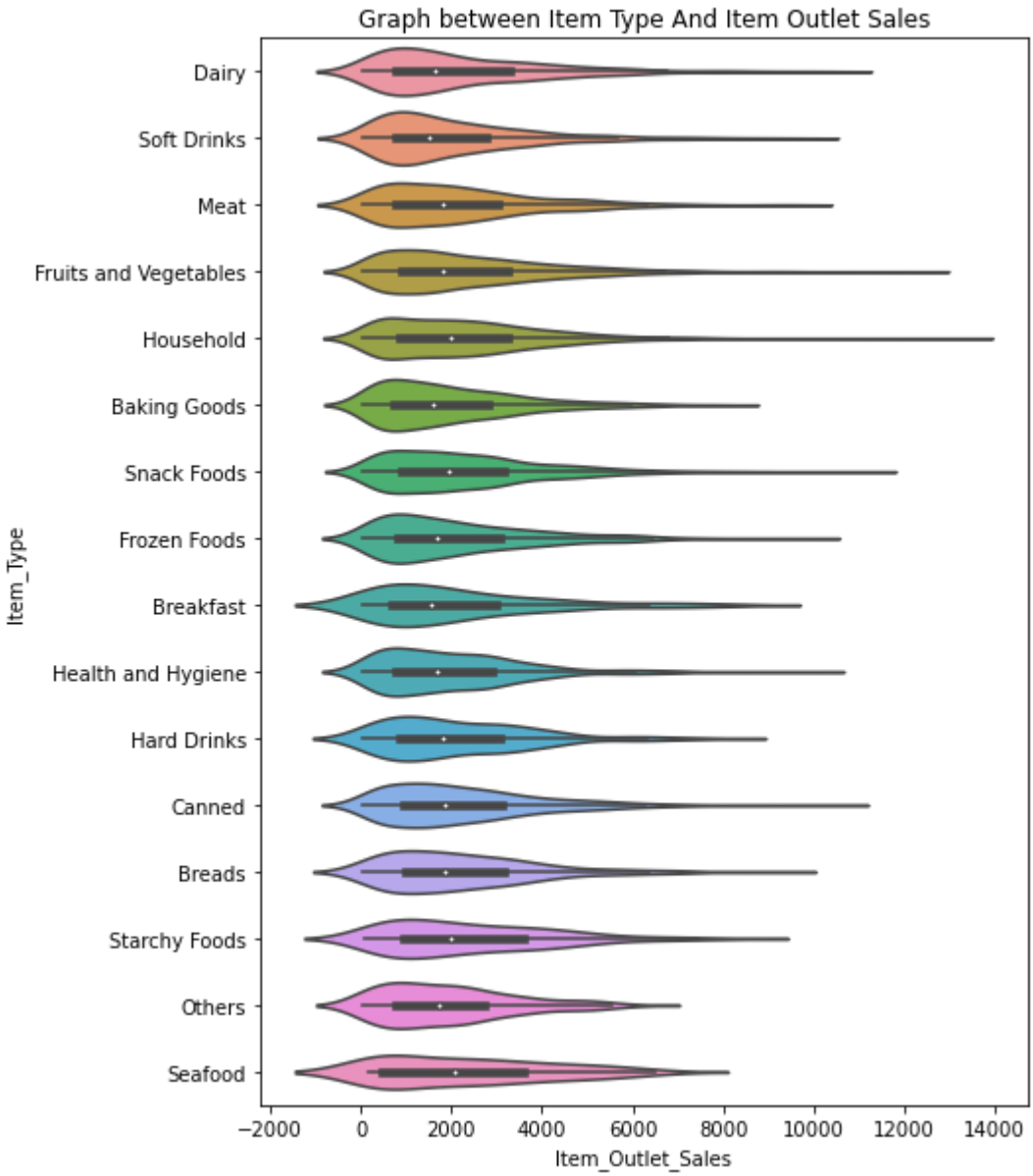
Out[23]:
```
'import sweetviz as sv\nmy_report = sv.analyze(data)\nreport = my_report.show_h
tml()'
```

2).DataVisualization

In [24]:
```
#countplot of Item_Fat_content
plt.figure(figsize=(6,4))
sns.countplot(x='Item_Fat_Content' , data=data,hue='Item_Fat_Content' )
plt.xlabel('Item_Fat_Content')
plt.title('countvalue of fat content')
plt.show()
```



In [25]:
```
#Graph between Item Type And Item Outlet Sales
plt.figure(figsize=(7,10))
sns.violinplot(data=data,x='Item_Outlet_Sales',y='Item_Type')
plt.title('Graph between Item Type And Item Outlet Sales')
plt.show()
```



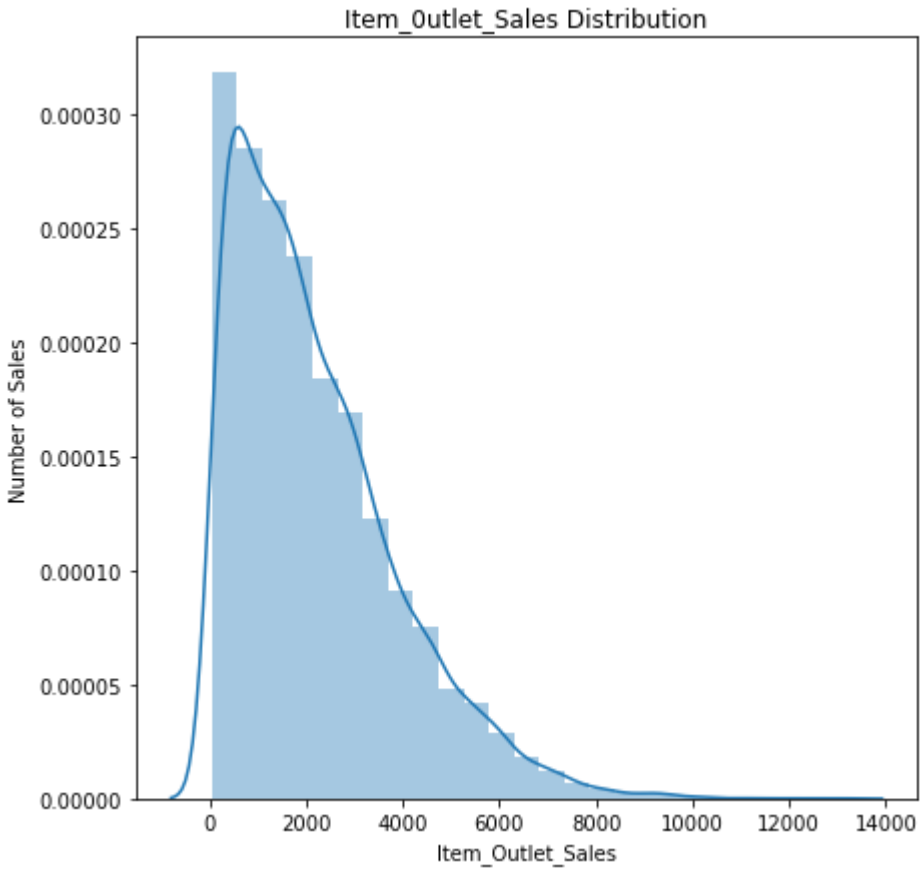
*from the above graph we conclude that

*household type things were highly sold out

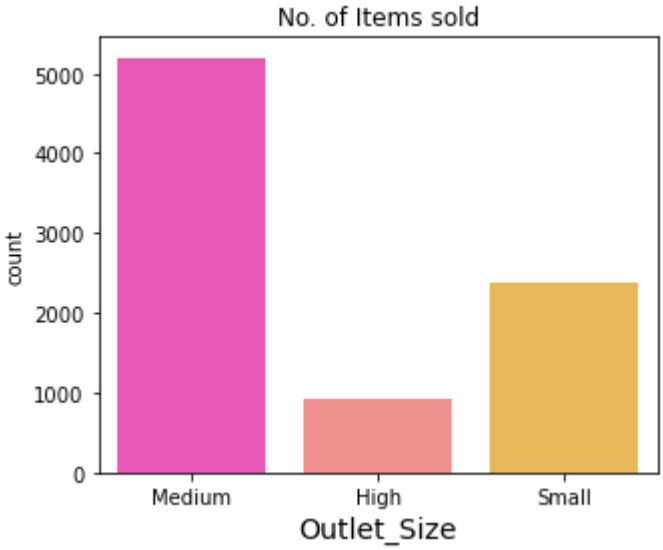
distribution of the variable outlet_size

In [26]:
```
plt.figure(figsize=(7,7))
sns.distplot(data.Item_Outlet_Sales,bins=25)
plt.xlabel("Item_Outlet_Sales")
plt.ylabel("Number of Sales")
plt.title("Item_0utlet_Sales Distribution")
```

Out[26]:
```
Text(0.5, 1.0, 'Item_0utlet_Sales Distribution')
```



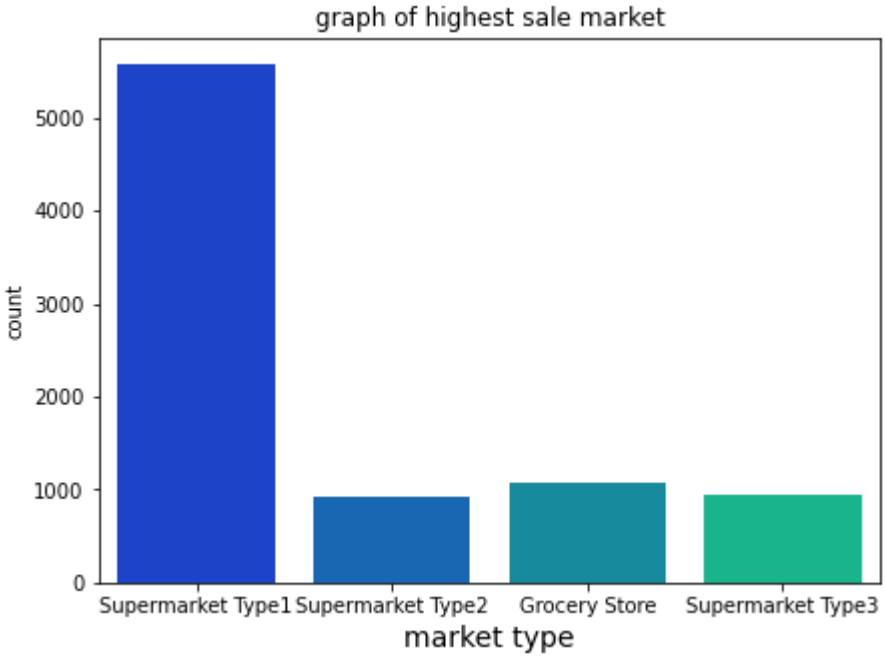
In [27]:
```
#showing the size of items
plt.figure(figsize=(5,4))
sns.countplot(x='Outlet_Size' , data=data,palette='spring')
plt.xlabel('Outlet_Size', fontsize=14)
plt.title('No. of Items sold')
plt.show()
```



Medium Sized Items--Highly Sold

graph of highest sale value

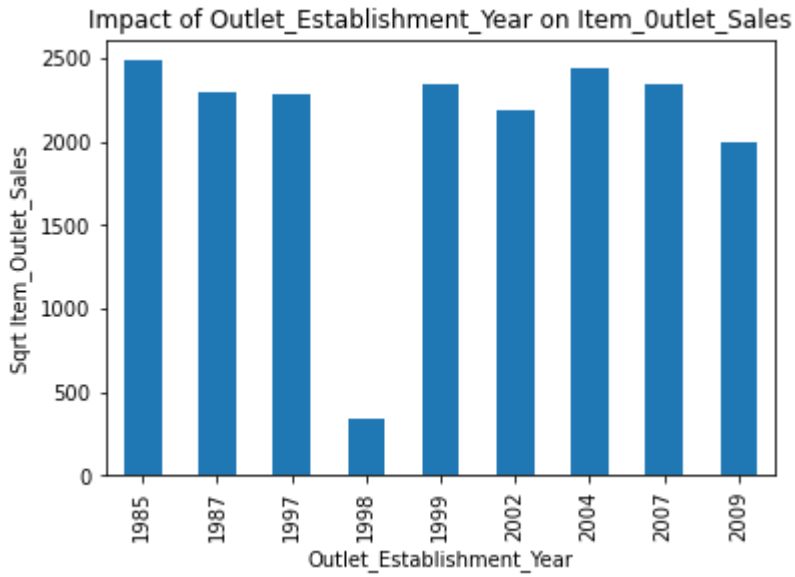
In [28]:
```
#graph of highest sale value
plt.figure(figsize=(7,5))
sns.countplot(x='Outlet_Type', data=data,palette='winter')
plt.xlabel('market type', fontsize=14)
plt.title('graph of highest sale market')
plt.show()
```



from the above graph we conclude that Supermarket Type 1 is the highest outlet type market

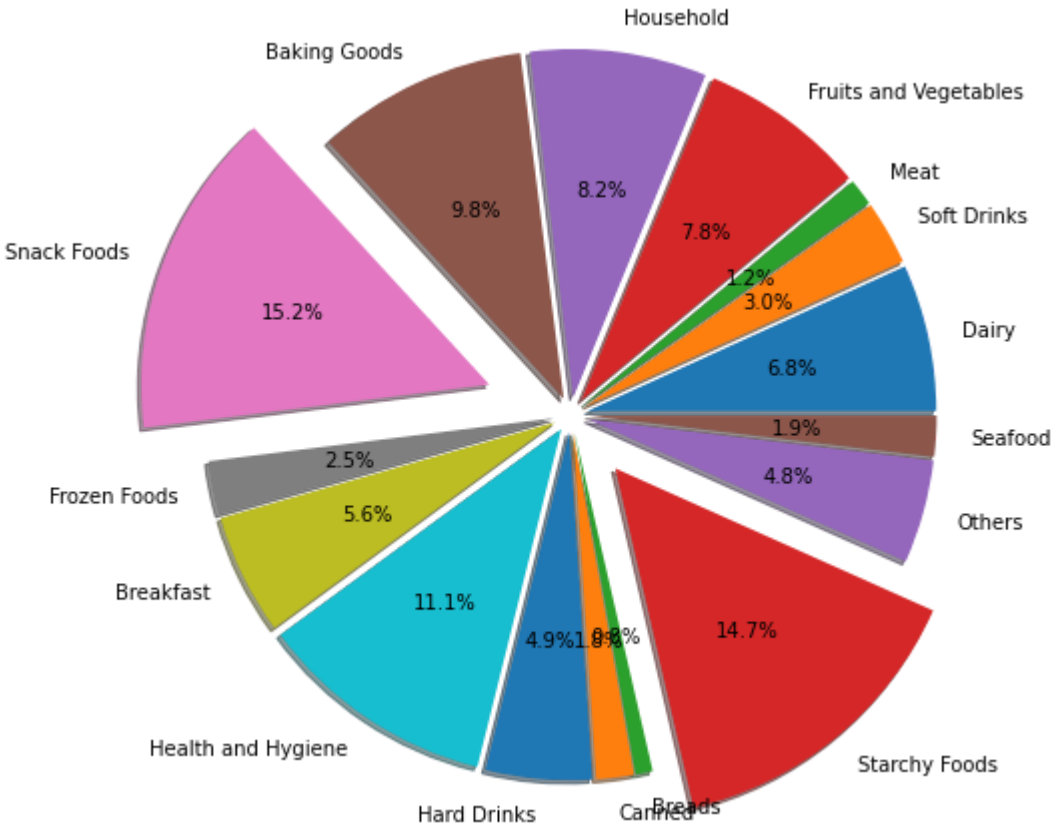
Outlet Establishment Year and Item Outlet Sales

In [29]:
```
data.groupby('Outlet_Establishment_Year')['Item_Outlet_Sales'].mean().plot.bar()
plt.xlabel("Outlet_Establishment_Year")
plt.ylabel("Sqrt Item_Outlet_Sales")
plt.title("Impact of Outlet_Establishment_Year on Item_0utlet_Sales")
plt.show()
```



pie chart that shows the item_Type & Item_Outlet_Sales

In [30]:
```
#shows the sale of Item_Type in percentage
fm1=data.groupby(by='Item_Type').sum()
fm=fm1['Item_Outlet_Sales']
plt.pie(fm, autopct = '%0.1f%%', radius = 2.0,
labels =['Dairy', 'Soft Drinks', 'Meat', 'Fruits and Vegetables',
       'Household', 'Baking Goods', 'Snack Foods', 'Frozen Foods',
       'Breakfast', 'Health and Hygiene', 'Hard Drinks', 'Canned',
       'Breads', 'Starchy Foods', 'Others', 'Seafood'],
        explode = [0.1,0.1,0.1,0.1,0.1,0.1,0.5,0.1,0.1,
                   0.1,0.1,0.1,0.1,0.4,0.1,0.1],shadow=True)
plt.show()
```



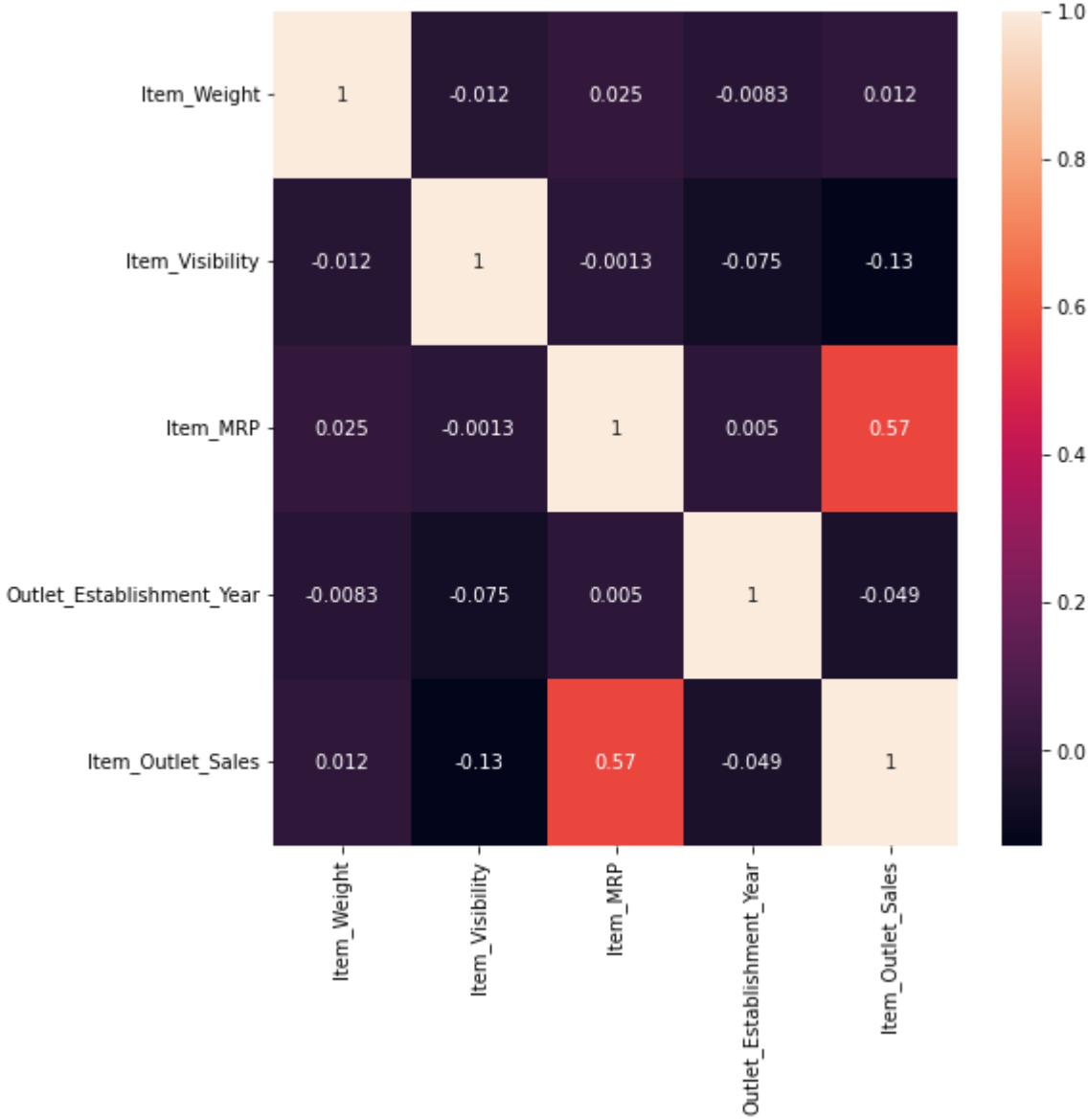
correlation

In [31]:
```
data.corr()
```

Out[31]:

	Item_Weight	Item_Visibility	Item_MRP	Outlet_Establishment_Year	Item_
Item_Weight	1.000000	-0.012049	0.024756	-0.008301	
Item_Visibility	-0.012049	1.000000	-0.001315	-0.074834	
Item_MRP	0.024756	-0.001315	1.000000	0.005020	
Outlet_Establishment_Year	-0.008301	-0.074834	0.005020	1.000000	
Item_Outlet_Sales	0.011550	-0.128625	0.567574	-0.049135	

In [32]:
```
#data correlation
plt.figure(figsize=(8,8))
sns.heatmap(data.corr(),annot=True)
plt.show()
```



selecting features based on general requirement

In [33]:
```
data.drop(['Item_Identifier','Outlet_Identifier'],axis=1,inplace=True)
```

In [ ]:

3).preprocessing task before model building

a) label encodding

In [34]:
```
from sklearn.preprocessing import LabelEncoder
```

In [35]:
```
le= LabelEncoder()
```

In [36]:
```
data=data.apply(le.fit_transform)
```

In [37]:
```
data
```

Out[37]:

	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Establishment_Ye
0	284	0	664	4	5592	
1	57	1	880	14	473	
2	376	0	715	10	2901	
3	393	1	0	6	4227	
4	265	0	0	9	627	
...	...	...	...	...	...	
8518	125	0	3912	13	4955	
8519	233	1	3278	0	2023	
8520	299	0	2302	8	1263	
8521	149	1	7175	13	1857	
8522	347	0	3108	14	1011	

8523 rows × 10 columns

b) splitting our data into train and test

In [38]:
```
from sklearn.model_selection import train_test_split
```

In [39]:
```
x=data.drop('Item_Outlet_Sales',axis=1)
```

In [40]:
```
#showning columns name of x
x.columns
```

Out[40]:
```
Index(['Item_Weight', 'Item_Fat_Content', 'Item_Visibility', 'Item_Type',
       'Item_MRP', 'Outlet_Establishment_Year', 'Outlet_Size',
       'Outlet_Location_Type', 'Outlet_Type'],
      dtype='object')
```

In [41]:
```
x.shape
```

Out[41]:
```
(8523, 9)
```

In [42]:
```
y=data['Item_Outlet_Sales']
```

In [43]:
```
y.shape
```

Out[43]: (8523,)

In [44]:
```
x_train,x_test,y_train,y_test=train_test_split(x,y,random_state=3,test_size=0.2)
```

In [45]:
```
data.shape
```

Out[45]: (8523, 10)

In [46]:
```
y_test.shape
```

Out[46]: (1705,)

c) standarization

In [47]:
```
#you see that you data is not standerdize
x_train.describe()
```

Out[47]:

	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Establishme
count	6818.000000	6818.000000	6818.000000	6818.000000	6818.000000	6818
mean	298.148724	0.353036	3715.789088	7.242006	2958.314022	3
std	95.863604	0.477949	2403.836586	4.211071	1674.212731	2
min	0.000000	0.000000	0.000000	0.000000	0.000000	0
25%	284.000000	0.000000	1594.500000	4.000000	1556.500000	1
50%	326.000000	0.000000	3718.000000	6.000000	2981.500000	4
75%	361.000000	1.000000	5814.750000	10.000000	4375.000000	6
max	415.000000	1.000000	7879.000000	15.000000	5937.000000	8

In [48]:
```
#standerdization is used to noralize our data
from sklearn.preprocessing import StandardScaler
sc= StandardScaler()
```

In [49]:
```
#we use fit_transform because we have to learn the data
x_train_std=sc.fit_transform(x_train)
```

In [50]:
```
#we use tranform because after learning we dont want data leakage
x_test_std=sc.transform(x_test)
```

In [51]:
```
x_train_std
```

Out[51]:
```
array([[ 0.24882254,  1.35372511,  0.47394838, ..., -0.28458515,
         1.0940992 ,  1.01648507],
       [ 1.16686078, -0.73870241, -0.68802946, ...,  1.38090972,
        -1.36455879, -0.25026602],
       [-0.10587405, -0.73870241,  0.70401417, ...,  1.38090972,
        -0.1352298 , -0.25026602],
       ...,
       [ 0.69740941, -0.73870241, -0.62021621, ..., -0.28458515,
         1.0940992 , -1.51701711],
       [ 0.29055155,  1.35372511,  1.19534816, ..., -0.28458515,
         1.0940992 ,  2.28323616],
       [-0.08500955, -0.73870241,  1.66671261, ..., -0.28458515,
         1.0940992 , -1.51701711]])
```

In [52]:
```
x_test_std
```

Out[52]:
```
array([[ 1.02080924, -0.73870241,  0.66865144, ..., -1.95008002,
         1.0940992 , -0.25026602],
       [ 0.22795803, -0.73870241,  0.83797653, ...,  1.38090972,
        -0.1352298 , -0.25026602],
       [-0.07457729, -0.73870241, -1.35825728, ..., -0.28458515,
         1.0940992 ,  1.01648507],
       ...,
       [-2.06713756, -0.73870241,  0.89538897, ...,  1.38090972,
        -1.36455879, -0.25026602],
       [ 0.29055155,  1.35372511, -1.1955887 , ..., -0.28458515,
         1.0940992 ,  2.28323616],
       [ 0.24882254, -0.73870241,  0.0629086 , ..., -0.28458515,
        -1.36455879, -0.25026602]])
```

In [53]:
```
y_train.head()
```

Out[53]:
```
7708    2951
4369     931
5978    2638
4117     229
5291    2633
Name: Item_Outlet_Sales, dtype: int64
```

In [54]:
```
y_test.head()
```

Out[54]:
```
4258    2100
2574    1948
517      358
3004     762
2159     145
Name: Item_Outlet_Sales, dtype: int64
```

d) model building

In [55]:
```
from sklearn.linear_model import LinearRegression
lr=LinearRegression()
```

In [56]:
```
lr.fit(x_train_std,y_train)
```

Out[56]:
```
LinearRegression()
```

In [57]:
```
y_pred_lr=lr.predict(x_test_std)
```

In [58]:
```
y_pred_lr.shape
```

Out[58]:
```
(1705,)
```

In [59]:
```
y_pred_lr
```

Out[59]:
```
array([1696.53375952, 1039.11532651, 1094.23390965, ..., 1969.71204   ,
       1699.77684825, 1033.55439761])
```

4) Evaluation using Linear regressor

In [60]:
```
from sklearn.metrics import r2_score,mean_absolute_error,mean_squared_error
```

In [61]:
```
print('mean_absolute_error:- ',mean_absolute_error(y_test,y_pred_lr))
print('r2_score:-            ',r2_score(y_test,y_pred_lr))
print('RMSE value:-          ',np.sqrt(mean_squared_error(y_test,y_pred_lr)))
```

```
mean_absolute_error:-  477.9858904633177
r2_score:-             0.5829985235005208
RMSE value:-           599.4950981614279
```

5) Evaluation using random forest regressor

In [62]:
```
from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor(random_state=3)
```

In [63]:
```
rf.fit(x_train,y_train)
```

Out[63]:
```
RandomForestRegressor(random_state=3)
```

In [64]:
```
y_pred_rf =rf.predict(x_test)
```

In [65]:
```
y_pred_rf
```

Out[65]:
```
array([2080.77, 1361.46,  680.88, ..., 2185.83, 1390.85,  768.96])
```

In [66]:
```
y_pred_rf.shape
```

Out[66]:
```
(1705,)
```

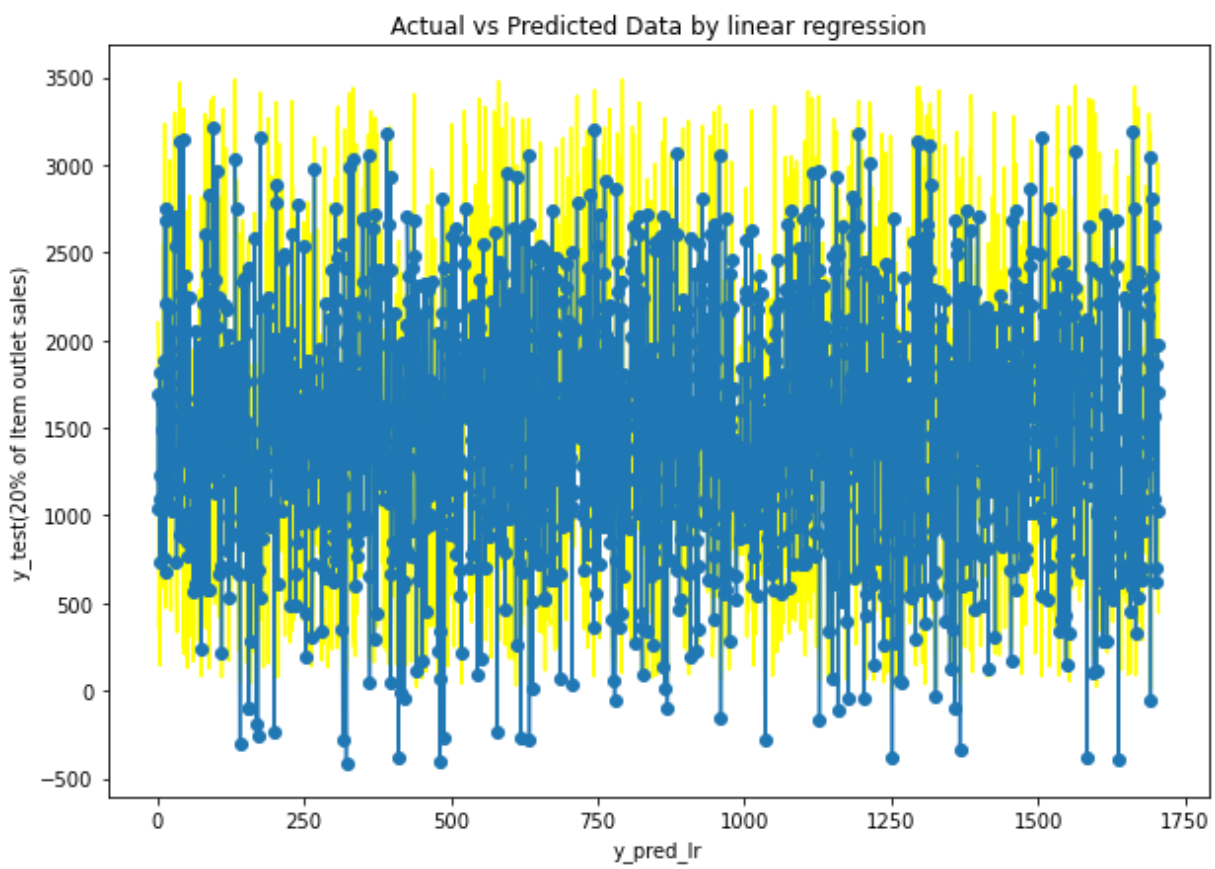
In [67]:
```
print('mean_absolute_error:- ',mean_absolute_error(y_test,y_pred_rf))
print('r2_score:-            ',r2_score(y_test,y_pred_rf))
print('Rmse value:-          ',np.sqrt(mean_squared_error(y_test,y_pred_rf)))
```

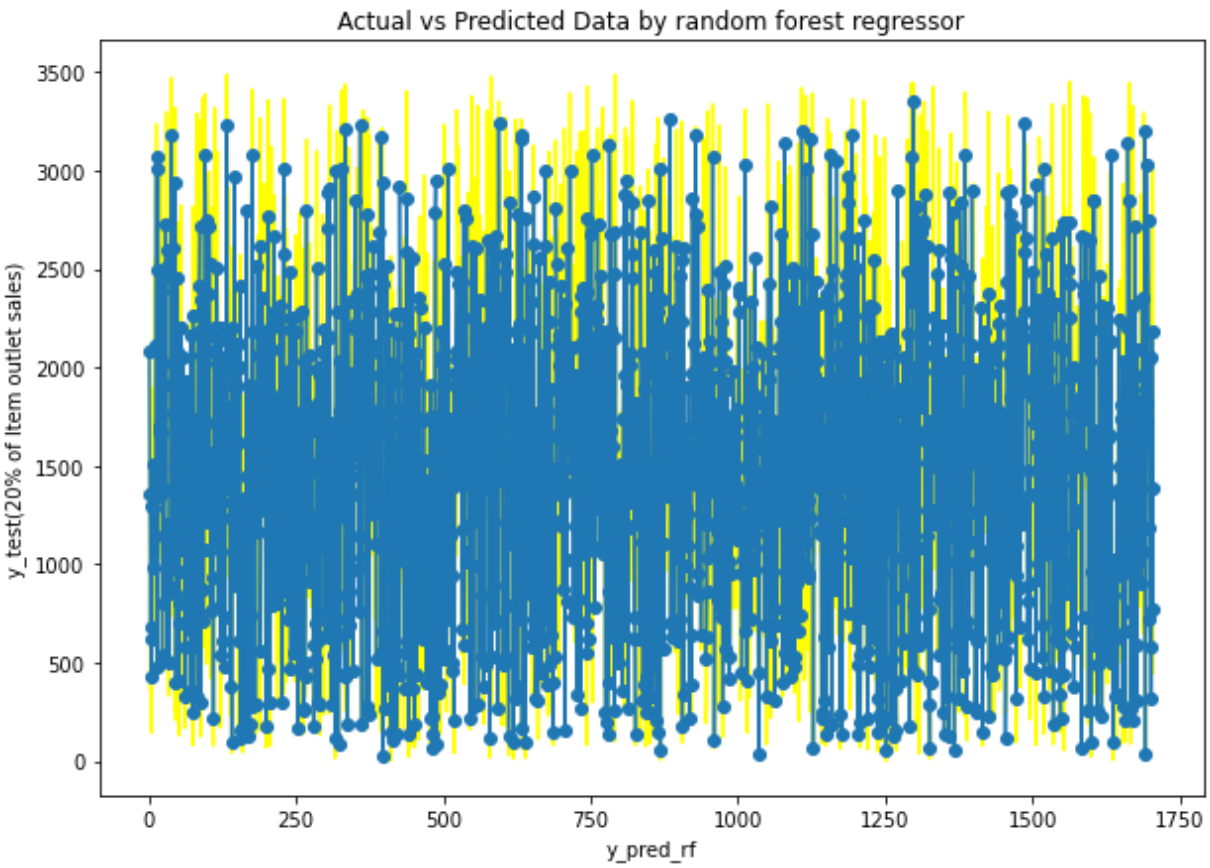
```
mean_absolute_error:-  417.9643988269795
r2_score:-             0.6536310037610501
Rmse value:-           546.369371481759
```

6) Actual vs Predicted sales graph

In [68]:
```
#graphh of prediction by using linear regression
plt.figure(figsize=(10,7))
c=[i for i in range(1,1706)]
plt.plot(c,y_test,color='yellow')
plt.plot(c,y_pred_lr,marker='o')
plt.xlabel('y_pred_lr')
plt.ylabel('y_test(20% of Item outlet sales)')
plt.title("Actual vs Predicted Data by linear regression")
```

Out[68]:
```
Text(0.5, 1.0, 'Actual vs Predicted Data by linear regression')
```



In [69]:
```
#graphh of prediction by using random forest regressor
plt.figure(figsize=(10,7))
c=[i for i in range(1,1706)]
plt.plot(c,y_test,color='yellow')
plt.plot(c,y_pred_rf,marker='o')
plt.xlabel('y_pred_rf')
plt.ylabel('y_test(20% of Item outlet sales)')
plt.title("Actual vs Predicted Data by random forest regressor")
```

Out[69]: Text(0.5, 1.0, 'Actual vs Predicted Data by random forest regressor')